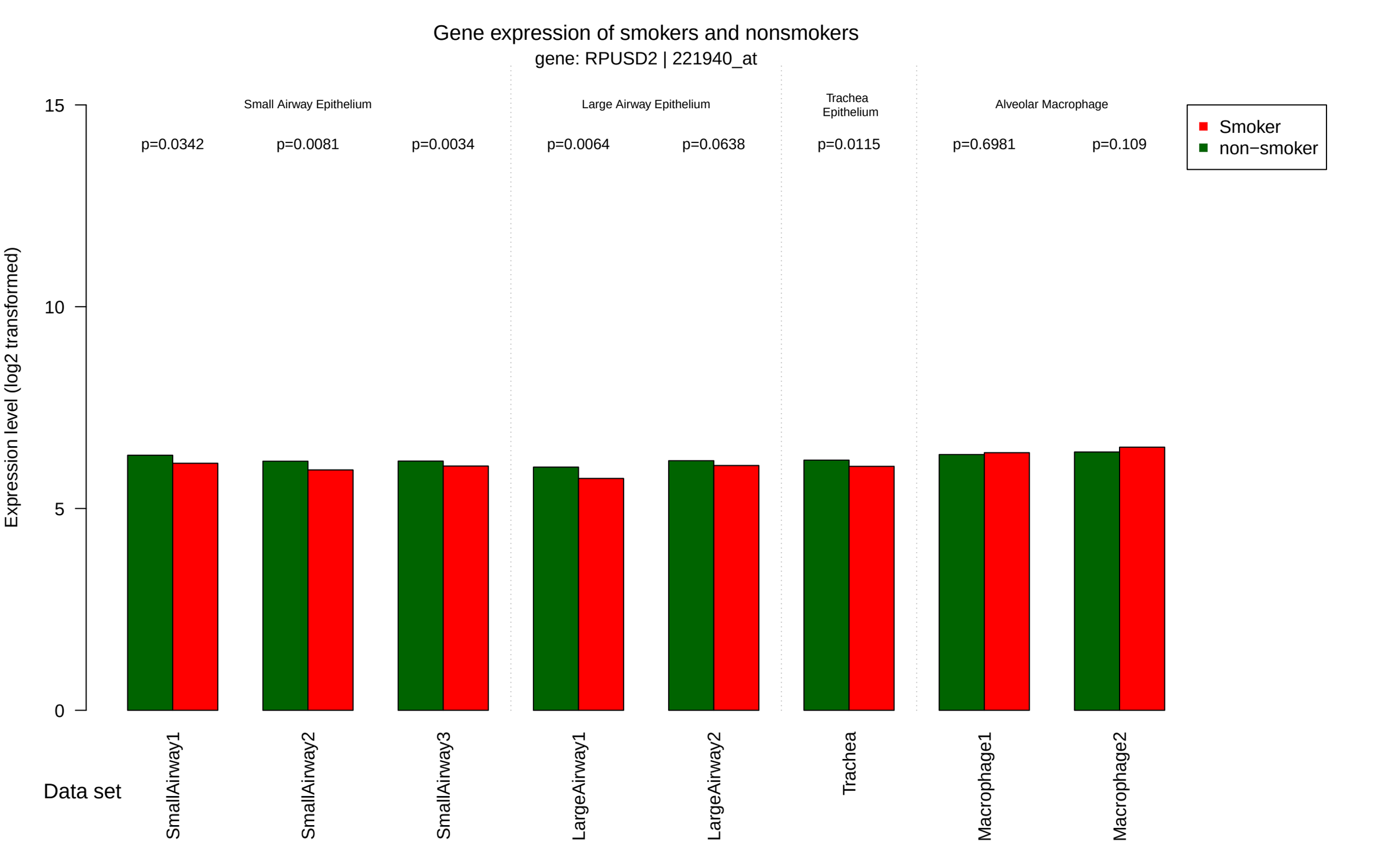
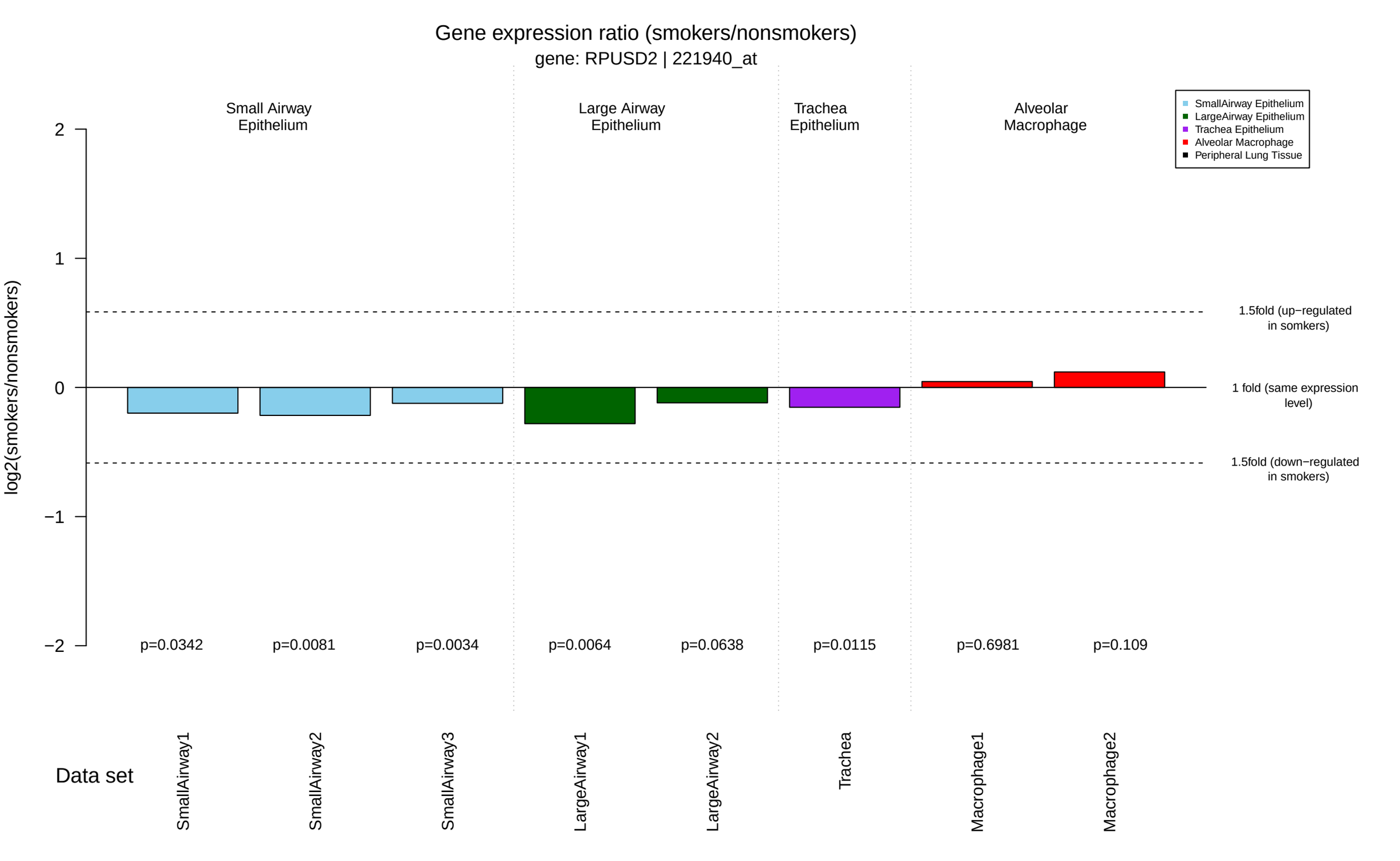
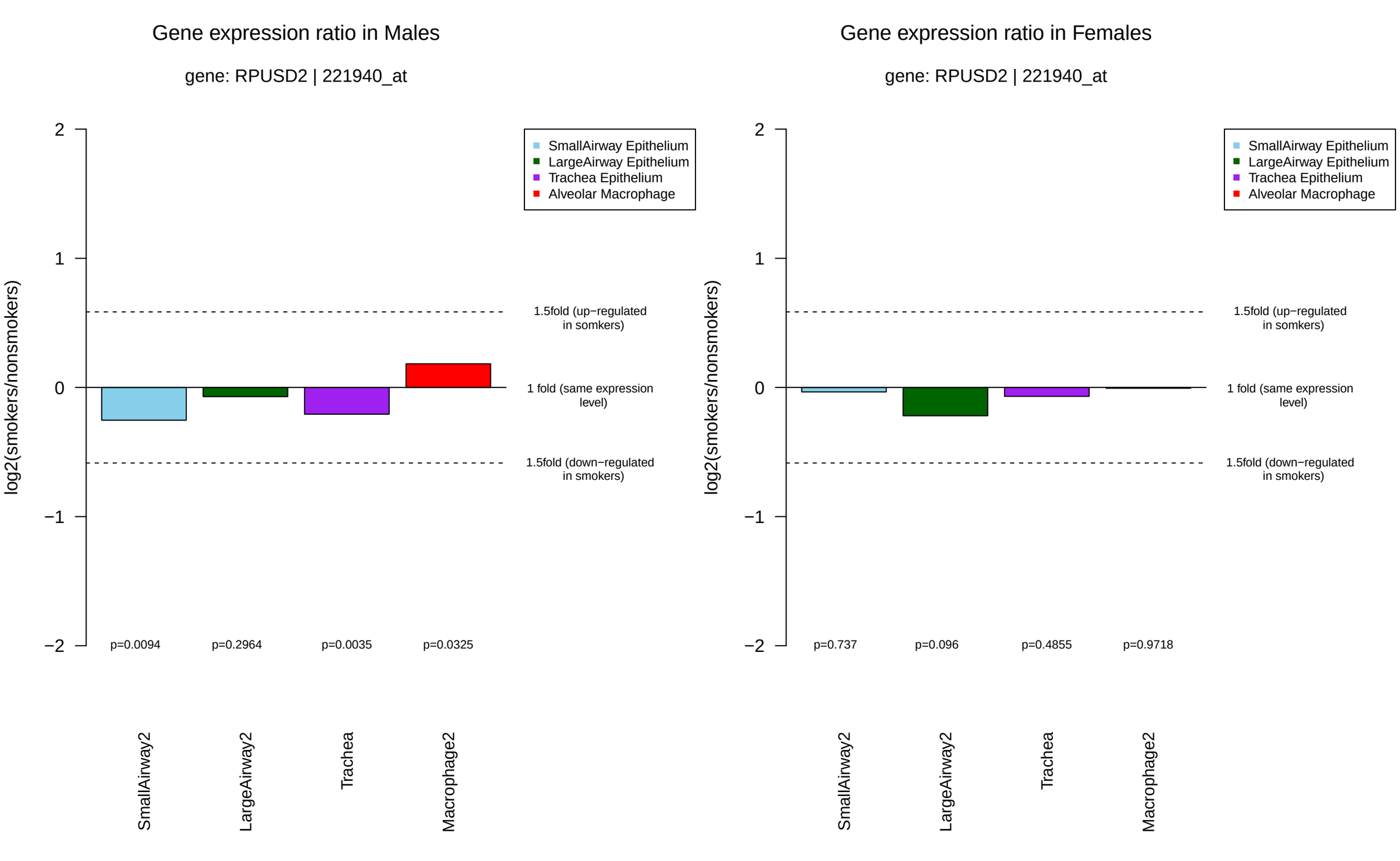
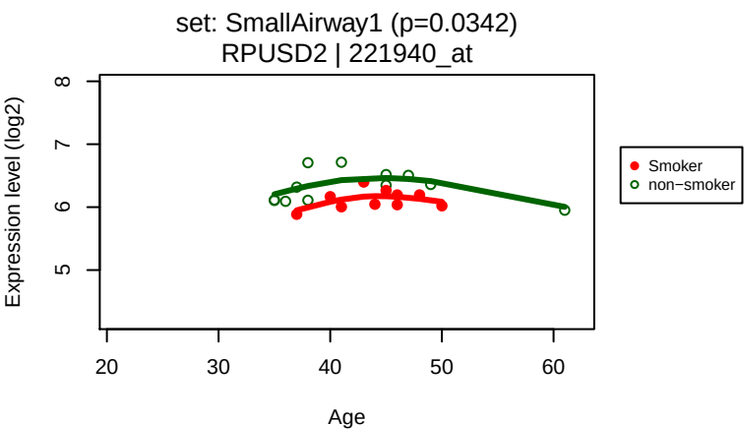
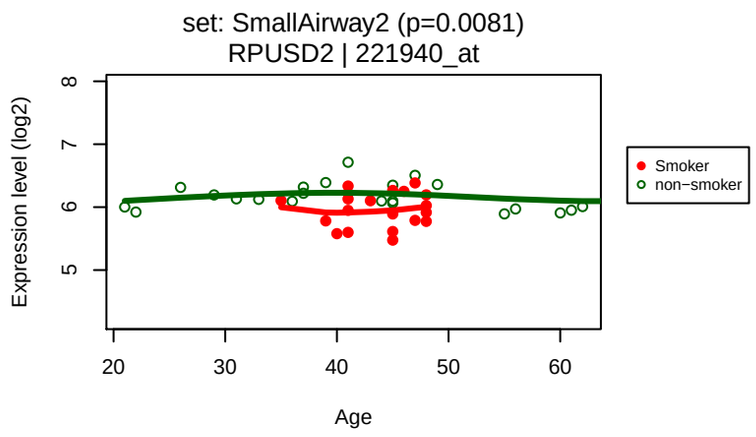
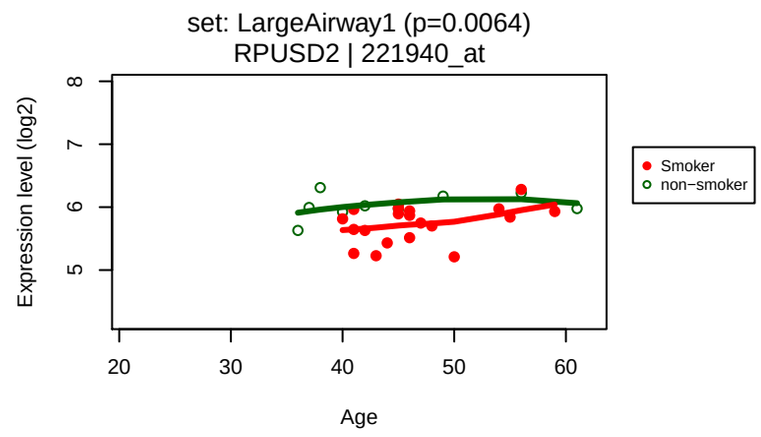

Gene expression of smokers and nonsmokers
gene: RPUSD2 | 221940_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0342
p=0.0081
p=0.0034
p=0.0064
p=0.0638
p=0.0115
p=0.6981
p=0.109
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

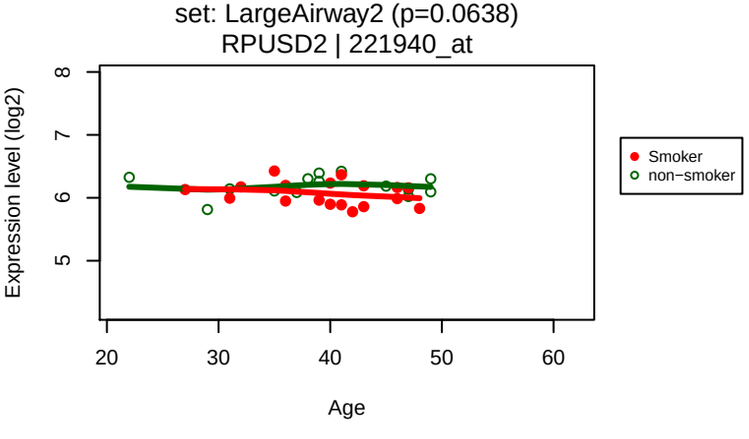

Gene expression ratio (smokers/nonsmokers)
gene: RPUSD2 | 221940_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0342
p=0.0081
p=0.0034
p=0.0064
p=0.0638
p=0.0115
p=0.6981
p=0.109
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

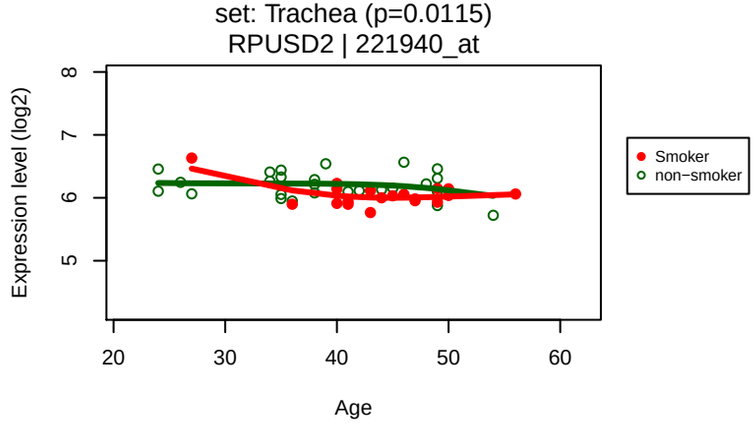

Gene expression ratio in Males
gene: RPUSD2 | 221940_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0094
p=0.2964
p=0.0035
p=0.0325
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPUSD2 | 221940_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.737
p=0.096
p=0.4855
p=0.9718
SmallAirway2
LargeAirway2
Trachea
Macrophage2

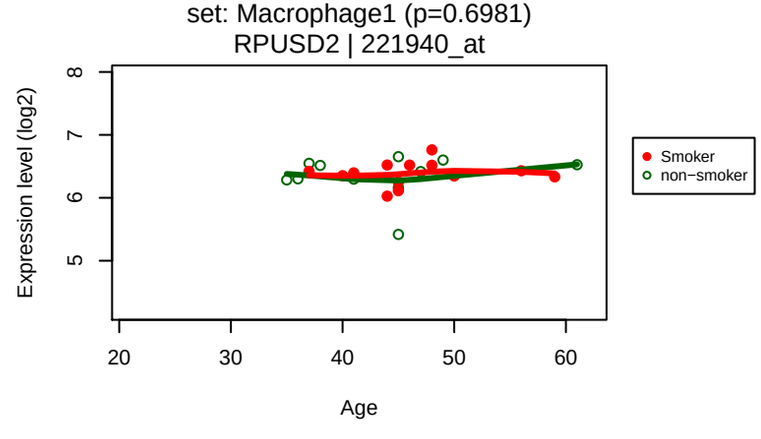

set: SmallAirway1 (p=0.0342)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker

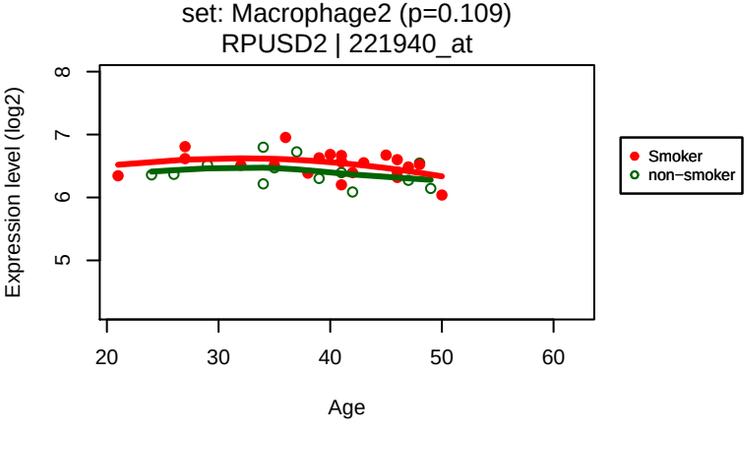

set: SmallAirway2 (p=0.0081)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker

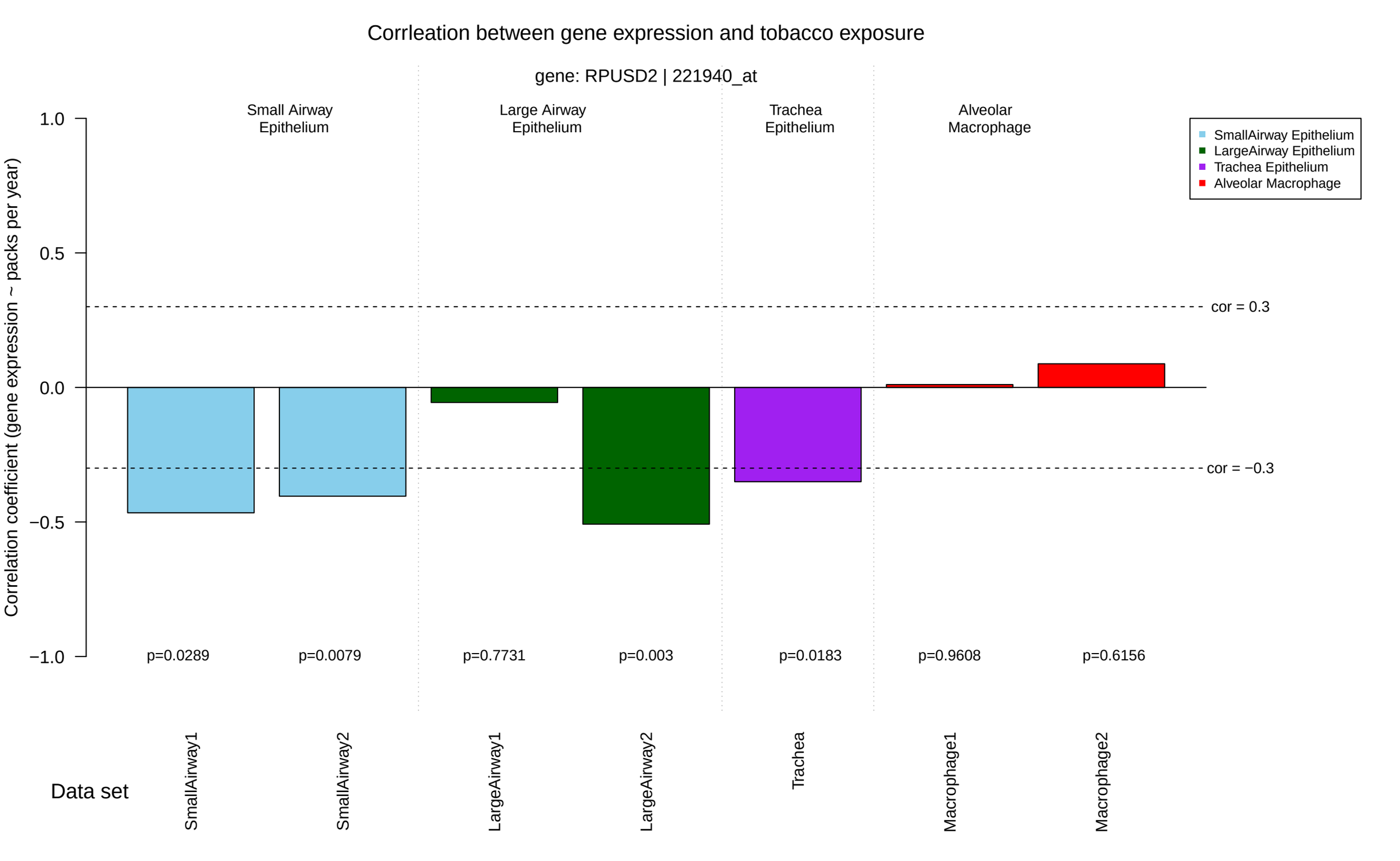

set: LargeAirway1 (p=0.0064)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0638)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0115)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6981)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.109)
RPUSD2 | 221940_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPUSD2 | 221940_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0289
p=0.0079
p=0.7731
p=0.003
p=0.0183
p=0.9608
p=0.6156
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2